

Multimedia Databases

Computing and Mathematics

May, 2026

Multimedia Databases (A14028)

Short Title:	Multimedia Databases	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Database and Analytics	
Author	JWOODS-OBRIEN	
Credits	5	Level: Advanced

Description of Module / Aims

This module will introduce the student to the principles and practice of designing distributed and object-oriented databases. The student will gain an understanding of multimedia database concepts, the architecture and design of a multimedia database. This module will also examine the procedures involved in the management and mining of multimedia databases.

Programmes

COMP-0633	BSc (Hons) in Creative Computing (WD_KCRC0_B)	4 / 8 / M
COMP-0633	BSc (Hons) in Multimedia Applications Development (WD_KMULM_B)	4 / 2 / M

Indicative Content

- Advanced Database Concepts: Distributed, Object-Oriented
- Multimedia Data & Metadata
- Modeling Multimedia (MM) Databases: Architectures, Information Models
- Managing MM Databases: Query Processing, Storage Management
- Mining MM Databases: Technologies & Techniques, Mining MM Data
- Management of Text, Image, and Video Databases

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the concepts, standards, and systems relating to distributed and object-oriented databases.
2. Evaluate the semantic nature of multimedia data, classify, generate and extract metadata for multimedia data types.
3. Determine the requirements and structures for the design, implementation and management of a multimedia database application.
4. Critique technologies and techniques appropriate to mining a multimedia database.
5. Evaluate statistical methods for text analysis, appropriate technologies for image processing and moving images.
6. Design and implement a MM database for a business scenario.

Learning and Teaching Methods

- The lectures will introduce the theory content to the student. The student will be encouraged to participate in class discussions and ask questions to support their learning process.
- The practical classes facilitate the student in implementing the theory learned in the lectures which in turn will form the continuous assessment.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4, 5
Continuous Assessment	50%	
<i>Project</i>	50%	2, 6

Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of distributed, object-oriented and multimedia databases.
- 40%-49%:** Be able to interpret and describe key concepts of distributed, object-oriented and multimedia databases.
- 50%-59%:** Ability to discuss key concepts of the design, management and mining of multimedia databases and have the ability to discover and integrate related knowledge in other knowledge domains.
- 60%-69%:** Be able to solve problems within the design, management and mining of multimedia databases by experimenting with the appropriate skills and tools.
- 70%-100%:** All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Connolly, T. and C. Begg. _Database Systems: A Practical Approach to Design, Implementation and Management_. 6th ed.. NY: Addison-Wesley, 2015.
- Dunkley, L. _Multimedia Databases: An Object Relational Approach_. UK: Pearson Education, 2003.

Requested Resources

- Room Type: Computer Lab